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Bridging accuracy and efficiency: Advancing mean radiant temperature measurement in Urban Ecology

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Abstract

Extreme summertime heat is an increasing challenge for cities, highlighting the need for accurate, spatially meaningful methods to measure and map heat in ways that reflect human thermal experiences and inform land management decisions. Mean radiant temperature (T_{mrt}) is a key metric for assessing urban heat at hyper-local scales, yet its measurement remains technically challenging. In this study, we apply the six-directional, gold standard method for measuring T_{mrt} with globe thermometer-based approaches across multiple levels of spatial aggregation and develop a novel machine learning model trained on field data. Data were collected in a semi-arid city in Colorado, USA, over two summers. Using measurements from residential parcels, we show that aggregated globe thermometer data—collected using a low-cost, accessible sensor—can capture thermal patterns across landscapes with reasonable accuracy. Our findings also indicate that machine learning, combining six-directional and globe thermometer data, offers promising potential for improving both measurement accuracy and efficiency. These findings are particularly relevant for planners working at the scale of parcels, where heat adaptation strategies are commonly applied, and especially insightful for semi-arid cities and those increasingly experiencing arid summer conditions due to climate change. This work advances practical methods for integrating human thermal comfort into landscape planning for climate-resilient urban design.

Introduction

Extreme weather and shifting climate regimes are increasingly impacting cities, challenging practitioners to understand how humans, flora, and fauna are experiencing temperature as they live and move across urban landscapes. Mean radiant temperature (T_{mrt})—the weighted total of all incoming shortwave and longwave radiation for a reference individual or object—is emerging as a key tool for capturing hyper-local thermal dynamics (ISO 7726 1998). T_{mrt} is especially useful for detecting and measuring fine-scale spatial and thermal heterogeneity, providing valuable information about how humans and other organisms experience heat in a city (Middel et al. 2019; 2021; Turner et al. 2022). This information can help inform strategies that promote human thermal comfort, manage heat exposure, and shape climate-responsive urban design.

Despite its growing importance, obtaining T_{mrt} in the field remains technically challenging. Scientists often rely on radiation modeling software like RayMan (Matzarakis et al. 2007; 2010), SOLWEIG (Lindberg et al. 2008), TUF-Pedestrian (Lachapelle et al. 2022), and ENVI-met (Bruse and Fleer 1998), which estimate T_{mrt} based on assumptions about urban geometry and biophysical processes. While these models offer useful estimates of radiative loading, they must be validated by on-the-ground measurements to ensure accurate microclimate assessments across diverse urban landscapes (Crank et al. 2020). Despite advancements in T_{mrt} measurement methods, the most accurate and reliable methods remain bulky, costly, and time-consuming to build (Rykaczewski et al. 2024). Thus, scientists

and practitioners are challenged to determine when less resource-intensive methods suffice for their research and management objectives and how to implement them effectively.

The highest cost, most accurate method for measuring T_{mrt} to date is often referred to as the six-directional method, integral radiation measurement, or gold standard approach- referred to hereafter as the six-directional method. This approach requires measuring shortwave and longwave radiation upward, downward, and in four orthogonal horizontal directions to estimate the average incoming radiation for a typical standing human (Höppe 1992; Thorsson et al. 2007; Middel and Krayenhoff 2019; Aleksandrowicz and Pearlmutter, 2023; Cheung et al. 2024). The setup typically involves mounting 12 radiometers on two extended arms at 1.1 m height, making them rather difficult to build, maneuver, and mass-produce. Six-directional T_{mrt} measurements have been shown to vary drastically with changes to ground and shade cover (Turner et al. 2022) and be strong predictors of spatio-temporal variation in human thermal comfort (Middel and Krayenhoff 2019), making the approach a valuable method for estimating the heat load on the human body.

A popular alternative to the six-directional approach is the globe thermometer method, which estimates T_{mrt} using air temperature (T_a), wind speed (V_a), and globe temperature (T_g)- the internal temperature of a small globe sensor. This widely used method is easier to implement, but the conventional formula for obtaining T_{mrt} (ISO 7726 1998) using the globe thermometer method yields results that differ from the six-directional approach. Discrepancies arise because T_g relies on convection, which, in turn depends on accurate V_a measurements to correct for. Due to these inaccuracies, researchers have sought to fine-tune the globe method equation to varying environmental conditions and globe sensor types by developing alternative formulas (Lin and Matzarakis 2011; Chen et al. 2014; Vanos et al. 2021; Ouyang et al. 2022; Liu et al. 2022). All methods are developed under specific environmental conditions and may or may not apply to other geographic or temporal settings.

Effective climate adaptation and urban heat mitigation depend on cities' ability to measure and monitor thermal dynamics at hyper-local scales that reflect the human experience. To better understand how people experience temperature at the ground level and how thermal conditions affect urban trees and other non-human organisms, scientists must find more accessible measurement techniques that provide accurate results. This study addresses the need for practical and ecologically informative approaches to T_{mrt} measurement that can support planning and policy decisions. Using a residential neighborhood in a semi-arid city as a test case, we evaluate the performance of low-cost globe thermometer readings (from a Kestrel 5400) compared to the six-directional, gold standard method (using a mobile biometeorological system called MaRTa). We used different globe thermometer conversion techniques from the literature and a novel machine learning approach. Specifically, we aim to 1) compare multiple Kestrel T_{mrt} methods to MaRTa across different levels of spatial aggregation, 2) assess how varying radiation components influence discrepancies between MaRTa and Kestrel T_{mrt} values for each of the methods, and 3) evaluate the impact of wind speed measurement accuracy on differences between the methods.

Our work is motivated by the hyper-local spatial scale at which planners and land managers operate and humans experience temperature; planning policies—from tree planting specifications to landscape design guidelines—are implemented at fine scales, and decision-makers require data that reflects how people experience heat within them. Our study is the first to assess the performance of various T_{mrt} methods in a semi-arid climate, and, while our fieldwork is grounded in a single city, the methodological insights apply to broader urban contexts, especially as cities in temperate zones begin to experience more arid summer conditions due to climate change.

Materials and Methods

Study Area

The study was conducted in the medium-sized city of Fort Collins, Colorado, USA, in July and August 2023 and 2024 (Fig. 1). Located at the base of the Rocky Mountains at 1,500 m (5,000 ft) elevation, Fort Collins experiences a semi-arid climate with 300 days of sunshine and 37 cm (15 in) of annual precipitation (City of Fort Collins). Data were collected on front yards within a 1.64 km^2 (0.63 mi^2) neighborhood in conjunction with other research examining thermal dynamics of varying front yard types (Benson et al., in press; Witvliet et al., unpublished results).

Weather Conditions

The first summer of data collection occurred during the region's wettest summer on record, while the second summer had more typical dry conditions (NOAA Online Weather Data 2023). Between June and August of 2023 (summer 1), the city experienced nearly three times the usual rainfall. The average summertime monthly rainfall from 1990 to 2022 was 44 mm, while in 2023, it was 129 mm per month. The minimum air temperature in the 2023 study period was 8 ℃, the maximum was 37 ℃, and the mean was 21 ℃. In 2024 (summer 2), the average summertime monthly rainfall was 25 mm, minimum air temperature was 8 ℃, maximum 39 ℃, and mean 22 ℃ (NOAA Online Weather Data 2023).

Sample Point Selection

We collected data on 32 front yards (Fig. 2) during the summers of 2023 and 2024, visiting yards in the morning and afternoon to capture diurnal variation. In 2023, each property was visited on two separate mornings and two afternoons at five points per yard. Morning measurements were taken at three properties sequentially at 08:30 h, 09:30 h, and 10:30 h, with the same properties revisited in the afternoon in the same order at 14:30 h, 15:30 h, and 16:30 h. During the first two weeks, houses were visited in a randomized order. Once all yards had been visited for a full day, the order was re-randomized for a second round of data collection. Data collection took place between July 17th and August 16th, 2023. In 2024, data collection followed a similar protocol between August 1st and August 14th, with a few modifications. Some houses from the 2023 sample were replaced with new ones, properties were

visited on a single day rather than two, and data were collected from four points within each yard instead of five. In total, 896 data points were collected across the two years.

Micrometeorology Measurements

A Kestrel 5400 meter and the biometeorology MaRTa cart collected micrometeorology data at each sample point on the front yards. The Kestrel meter was mounted on a tripod at a height of 1.1 meters to match the height of the MaRTa sensors, and the two devices were positioned opposite each other on either side of the sample point (Fig. 3). After allowing the Kestrel to acclimate to its environment at each measurement point, meteorological data were collected and averaged over one minute.

The MaRTa platform is equipped with 12 Apogee radiometers to measure shortwave and longwave radiation in six directions, the Campbell Scientific HygroVue10 for air temperature (T_a) and relative humidity housed within an Apogee TS-100 SS fan aspirated radiation shield, an Apogee Infrared Radiometer for surface temperature, and a Gill Ultrasonic Anemometer for measuring wind speed (V_a) (Fig. 4). Adapted from the MaRTy machine developed by Middel and Krayenhoff (2019), MaRTa calculates mean radiant temperature (T_{mrt}) using the following equation:

$$T_{mrt} = \sqrt[4]{\frac{\sum_{i=1}^{6} W_i (a_k K_i + a_l L_i)}{a_l * \sigma}} - 273.15\ K$$

1

Where K_i = incoming shortwave radiation, L_i = incoming longwave radiation, a_k = human emissivity for shortwave radiation (0.7), a_l = human emissivity for longwave radiation (0.97), σ = the Stefan-Boltzmann constant ($5.67 * 10^{-8}$), and W_i = weighting parameters for the angles of a standing human (0.06 for upward and downward radiation; 0.22 for lateral radiation) (Middel and Krayenhoff 2019).

Alongside the MaRTa cart, we used a Kestrel 5400, which features a 25 mm diameter black-coated copper globe. The micrometeorology metrics utilized from the Kestrel were globe temperature (T_g), (V_a), and T_a, which were used to estimate T_{mrt}.

We avoided taking micrometeorology measurements during rain events, defined as periods of active rainfall, to prevent data for Benson et al. (in press) from being highly skewed by unusual weather conditions. Given the exceptionally rainy conditions of the summer, measurements were occasionally taken during light drizzling.

Analysis

Objective 1: Compare multiple Kestrel mean radiant temperature ($\varvec{T}_{\varvec{mrt}}$) methods to MaRTa across different levels of spatial aggregation

By comparing to gold standard six-directional T_{mrt} values, we tested the accuracy of six methods for converting globe thermometer readings to T_{mrt} at the hyper-local level (Table 1). Each method's accuracy was determined by extracting RMSE for the Kestrel's T_{mrt} estimate compared to MaRTa's T_{mrt} readings. Additionally, minimum, maximum, mean, and variation from the mean were considered for each T_{mrt} method to understand the spread of data points. Of the Kestrel-based methods, five are from the literature and one utilizes a novel machine learning approach. All analyses were performed in RStudio (RStudio Team 2020) using the "tidyverse" package (Wickham et al. 2019).

Table 1

Summary of the globe thermometer methods for calculating mean radiant temperature (T_{mrt}) examined in this study, including location of the study and the type of globe thermometer(s) used, if available.

Authors	Location	Globe Thermometer: Color | Diameter
Thorsson et al. (2007)	Various	Gray | 38 mm
Vanos et al. (2021)	Tempe, Arizona, USA	Black | 150 mm diameter
Ouyang et al. (2022)	Hong Kong	Black | 40 mm Black | 25.4 mm Gray | 40 mm
Liu et al. (2022)	Tianjin, China	Black | 70 mm
Lin and Matzarakis (2011)	N/A	N/A
This study	Fort Collins, Colorado, USA	Black | 25 mm

The conventional method, as outlined by Thorsson et al. (2007), uses the following equation:

$$T_{mrt} = \left[\left(T_g + 273.15\right)^4 + \frac{1.1 * 10^8 * {V_a}^{0.6}}{\epsilon_g * D^{0.4}}\left(T_g - T_a\right)\right]^{0.25} - 273.15$$

2

Where T_g is globe temperature (°C), V_a is wind speed (m/s), T_a is air temperature (°C), ϵ_g is globe emissivity (0.95), and D is globe diameter. For the globe diameter, 150 mm was input because of internal conversions the Kestrel uses when calculating T_g.

The next method used a formula generated by Vanos et al. (2021) to correct for asymmetrical heating of globe thermometers. The corrected equation was developed under warm-hot and clear weather conditions in Tempe, Arizona, and resulted in the following:

$$T_{mrt} = \sqrt[4]{\left(\overline{T}_s + 273.15\right)^4 + \left(0.24 + 2.08{V_a}^{0.5} + 1.14{V_a}^{0.667}\right)\left(\overline{T}_s - T_a\right)10^8} - 273.15$$

3
Where,

$$\overline{T}_s = 1.345T_g - 0.369T_a + 0.725$$

Third, a method developed by Ouyang et al. (2022) was tested. The author's formula was calibrated to a humid subtropical climate in Hong Kong, resulting in the following:

$$T_{mrt} = \left[\left(T_g + 273.15\right)^4 + \frac{0.678 * 10^8 * {V_a}^{0.019}/D^{0.4}}{\epsilon_g}\left(T_g - T_a\right)\right]^{0.25} - 273.15$$

4
Fourth, a method outlined by Liu et al. (2022) was applied, where the authors used machine learning to alter the conventional formula, resulting in the following new equation:

$$T_{mrt} = \left[\left(T_g + 273.15\right)^4 + \frac{1.07 * 10^8 * {V_a}^{0.251}}{\epsilon_g * D^{0.4}}\left(T_g - T_a\right)\right]^{0.25} - 273.15$$

5
The fifth method tested was proposed by Lin and Matzarakis (2011), where a constant wind speed value of 0.5 m/s was input into the conventional formula (Chen et al. 2014).

Lastly, a random forest model was developed using a 70/30 split in the "randomForest" R package (Liaw and Wiener 2002); 70% of the MaRTa and Kestrel data pairs were randomly selected to train the model (n = 609), while the remaining 30% was used to test its accuracy (n = 258). The model used T_g, T_a, and V_a from the Kestrel as input variables to build an ensemble of decision trees that predict T_{mrt} measured by MaRTa.

In addition to inspecting individual points of Kestrel and MaRTa data, we also investigated the performance of globe thermometer methods at the front yard level by averaging all points per yard (five points of data were collected per yard in 2023, four points were collected in 2024). For each method of Kestrel-derived T_{mrt}, we aggregated points at the front yard level and then compared them to MaRTa aggregations at the same level by 1) comparing each method individually to MaRTa T_{mrt} and extracting RMSE, and 2) examining differences in averages and deviation from the mean between all methods. We excluded the random forest model from the front yard level analysis, as its 70/30 train-test split was applied across all points regardless of yard boundaries, resulting in incomplete or missing yard-level data that prevented consistent averaging.

Objective 2: Assess how varying radiation components influence discrepancies between MaRTa and Kestrel mean radiant temperature ($\varvec{T}_{\varvec{mrt}}$) values

Because high incoming solar radiation is known to skew globe thermometer data, we isolated MaRTa radiation components to examine impacts on ΔT_{mrt}. First, differences in T_{mrt} between MaRTa and Kestrel were calculated for each data point ($\Delta T_{mrt} = MaRTa\, T_{mrt} - Kestrel\, T_{mrt}$; n = 867). Then, for each method of Kestrel T_{mrt}, we examined relationships between total incoming shortwave and longwave radiation on ΔT_{mrt}. Total shortwave and longwave radiation were calculated as a weighted total according to the six-directional T_{mrt} equation.

Objective 3: Evaluate the impact of wind speed measurement accuracy on differences between the methods

We performed a sensitivity analysis on V_a measurement accuracy for the Kestrel-derived T_{mrt} methods to determine the amount of error introduced using a low-cost wind speed sensor. For each method, V_a readings from MaRTa's sonic anemometer were input instead of those from the Kestrel's impeller and change in RMSE was noted. The Lin and Matzarakis (2011) method was excluded from this analysis because it does not require V_a inputs from the Kestrel (it uses a stable wind speed value of 0.5 m/s). For the random forest model method utilizing T_g, T_a, $\text{and}\ V_a$ to predict MaRTa T_{mrt}, we replaced V_a with MaRTa V_a to see if that improved the model performance. We also extracted feature importance for each variable in both random forest model versions.

Results

Objective 1: Compare multiple Kestrel mean radiant temperature (T_{mrt}) methods to MaRTa across different levels of spatial aggregation

We calculated and compared multiple globe thermometer methods for estimating T_{mrt} at two levels of spatial aggregation to evaluate the efficacy of each method on our data and observe changes in discrepancies incurred by shifting the level of spatial aggregation. By comparing each Kestrel T_{mrt} method to the gold-standard, six-directional MaRTa machine, we found that the random forest model performed best at the single-point level, compared to the other methods selected from the literature (Fig. 5). Out of the literature-based methods, the Lin and Matzarkis (2011) method worked best on our data and the Vanos et al. (2021) method performed the poorest. Furthermore, discrepancies between Kestrel and MaRTa T_{mrt} decreased when examining T_{mrt} aggregated at the front yard level. Specifically, each of the methods decreased in RMSE when scaling up from the hyper-local, individual point level. Overall, the globe thermometer methods showed closer agreement with MaRTa T_{mrt} at lower ranges than at the higher ranges of T_{mrt}.

MaRTa showed less variation in T_{mrt} compared to the globe thermometer methods, apart from the machine learning method (Fig. 6). The Vanos et al. (2021) method had the most variation in values, while the random forest model had the least.

Objective 2: Assess how varying radiation components influence discrepancies between MaRTa and Kestrel mean radiant temperature ($\varvec{T}_{\varvec{mrt}}$) values

High solar radiation via sun exposure is known to cause errors in globe thermometer readings; we assessed the impact of shortwave radiation on our observed Kestrel T_{mrt} by deconstructing radiation components of MaRTa T_{mrt} and comparing that to Δ T_{mrt}, where Δ T_{mrt} = $MaRTa\,T_{mrt} - Kestrel\,T_{mrt}$. As total shortwave radiation increased, all six methods of Kestrel T_{mrt} increased in error, while incoming total longwave radiation appeared to have little influence on Δ T_{mrt} (Fig. 7).

Objective 3: Evaluate the impact of wind speed ($\varvec{V}_{a}$) measurement accuracy on differences between the methods

For each globe thermometer method estimating T_{mrt}, we investigated error reduction incurred from use using a low-cost V_a sensor and found a slight reduction in RMSE for most methods (Table 2). On average, replacing the Kestrel impeller with MaRTa's sonic anemometer resulted in a reduction in RMSE of 0.78. The random forest model, which uses a distinct approach from the other methods to determine T_{mrt}, showed little improvement from this alteration. This likely reflects the low influence of V_a in the model overall. Feature importance values—indicators of each variable's relative contributions—were low for both Kestrel V_a (7.55) and MaRTa V_a (5.74), where a lower value indicates less importance in the model. This is compared with T_g (68.06 for the Kestrel model, 77.93 for the MaRTa model) and T_a (32.81 and 31.28, respectively).

Table 2

Impact of using MaRTa's sonic anemometer in place of the Kestrel 5400's impeller for calculating T_{mrt} at the individual point level. The second column shows the change in RMSE associated with each method, while the third column shows the resulting RMSE for swapping in MaRTa wind speed (V_a) for each Kestrel method.

Method	**Difference in RMSE using MaRTa $\varvec{V}_{a}$**	**Resulting RMSE using MaRTa $\varvec{V}_{a}$**
Thorsson et al. (2007)	-0.87	8.72
Vanos et al. (2021)	-1.96	9.74
Ouyang et al. (2022)	-0.99	8.13
Liu et al. (2022)	-1.14	7.91
Lin & Matzarakis (2011)	--	--
Random Forest Model	+ 0.27	6.76

Discussion

Challenges to the Kestrel Approach: Wind Speed, Globe Type Solar Radiation

For conventional globe thermometer-based methods, it has long been known that wind speed (V_a) dramatically influences T_{mrt} calculations, as the conventional method was originally developed for indoor environments with minimal air movement (ISO 7726 1998). Our analysis showed that a simple and effective way to smooth out Kestrel T_{mrt} inaccuracies was by applying a stable, low V_a constant to the Thorsson et al. (2007) method, as recommended by Lin and Matzarakis (2011) and Chen et al. (2014). Applying V_a from the MaRTa machine's sonic anemometer to the various Kestrel T_{mrt} calculations also improved our models, but to a lesser degree than the Lin and Mazarakis (2011) method. This could be partially due to MaRTa's placement slightly away from the Kestrel, meaning the two instruments may have experienced distinct V_a.

In the case of the random forest model, performance did not improve with the use of MaRTa V_a, likely because V_a was not a particularly important feature to the model in the first place. Unlike the other methods, which are based on formulas where V_a directly influences the outcome, the random forest model uses data-driven decision trees and does not require any specific predefined relationships between inputs and the response. If a variable does not contribute substantially to improving predictive accuracy—either because it has limited variation or is less informative relative to other inputs—the model will assign it less importance. In this case, V_a contributed less than T_g and T_a, which more strongly predicted T_{mrt} across the dataset.

Another factor influencing our globe-based T_{mrt} results is the globe sensor type. Each of the methods we examined were developed using globes that vary in size, material, and color. The Kestrel 5400 used in this study features a 25 mm black-coated copper globe, which equilibrates to its environment more quickly than larger globes but is also more susceptible to convective error (Kántor et al. 2015). Notably, the Kestrel internally converts its T_g readings to simulate those of a standard 150 mm globe thermometer. We expect that some of the discrepancy between our results and those reported in other studies arises from mismatches in globe sensor types. For example, Vanos et al. (2021) developed a T_{mrt} equation using data from a 150 mm globe. While the Kestrel's output is adjusted to match this standard, its actual globe is much smaller. Therefore, we expect that had we used a physical 150 mm globe, the results from our Vanos et al. (2021) test would have more closely aligned with the six-directional T_{mrt}.

The radiation decomposition analysis (Objective 3) added to our understanding of observed Kestrel T_{mrt} error, indicating that the globe thermometer tended to overpredict T_{mrt} under conditions of high shortwave radiation, which is consistent with Chen et al. (2014) and Liu et al. 2022, but contrary to Thorsson et al. (2007). This discrepancy is likely due to our and Chen et al.'s (2014) use of a black globe sensor, which may "overheat" and measure hotter temperatures, whereas Thorsson et al. used a gray

globe, which is closer in albedo to the human skin. This implies that solar radiation drives differences between globe thermometer and six-directional T_{mrt}, suggesting that the black globe thermometer readings become unstable under conditions with high shortwave radiation.

Spatial Aggregation of Globe Thermometer-Derived Mean Radiant Temperature Reveals Thermal Patterns in Heterogeneous Landscapes

Our results show that Kestrel-derived (globe thermometer) T_{mrt} aligns more closely with MaRTa (the gold standard for T_{mrt} measurements) when averaging across four to five points per yard, compared to its performance at the single-point level. RMSE values between Kestrel and MaRTa T_{mrt} were lower at the yard level, while hyper-local (single point) discrepancies were more pronounced. The front yards we studied ranged from 22 to 246 m² in area—roughly the size of a pocket park or green roof. These findings suggest that, in the absence of access to a six-directional platform, averaging multiple measurements across small urban spaces can capture broad thermal patterns with reasonable accuracy.

The focus of urban ecologists is understanding how ecosystem processes interact with landscape heterogeneity, rather than isolated data points. Our findings indicate that averaging Kestrel T_{mrt} yields a practical and reliable approach for microclimate assessment in front yards, even if the Kestrel struggles to pinpoint extremely fine-scale variation. Indeed, Benson et al. (in press), using the same dataset, shows that aggregated T_{mrt} measurements detect landscape thermal patterns where air temperature (T_a) lacks sensitivity. These findings highlight that, despite its limitations, the Kestrel can effectively link urban landscape structure to biophysical outcomes when spatially aggregating.

In addition to spatial aggregation and other previously discussed factors influencing the accuracy of the globe thermometer method, another relevant consideration is the temporal resolution of T_{mrt} data. Thorsson et al. (2007) recommended averaging globe thermometer readings over five-minute periods to reduce inconsistences. In our study, we recorded data at one-minute intervals, consecutively across the yard, such that the final T_{mrt} value for each front yard reflects a five-minute average. Thus, our method involves both temporal and spatial aggregation: the former through consecutive point measurements, and the latter through sampling across multiple locations within a yard. While we acknowledge that both types of aggregation contribute to error reduction, we expect spatial aggregation to be the dominant factor. This is because T_{mrt} variation across space is likely greater than variation across the short time span needed to complete measurements on a single yard. That being said, we cannot fully disentangle the contributions of spatial versus temporal averaging in this design, and the role of temporal smoothing warrants further attention in future work.

Machine Learning for Refined Mean Radiant Temperature Field Methods

Ours is the first study to incorporate a machine learning technique to predict T_{mrt} based on Kestrel data. By using MaRTa (six-directional) as a highly-accurate T_{mrt} reference, we were able to evaluate the

performance of Kestrel metrics and found that this approach holds promise in bolstering Kestrel-based field methods. However, more data across diverse landscapes and under a range of weather conditions would be needed to develop a robust model for use beyond this study. Localized research may benefit from pursuing tailored machine learning models developed under a range of local weather and landscape conditions. Although this approach still relies on a six-directional machine for model training, its role shifts from being the primary T_{mrt} data source to serving as a tool for building models at key moments during a field campaign.

We suspect that our random forest model would be further strengthened by incorporating shade into the model. In machine learning models built to predict human thermal experience, Guo et al. (2024) split their micrometeorology data into two sets based on presence of shade, while AlKhaled et al. (2024) included shade presence as a variable in their models to predict T_{mrt} based on air temperature and a variety of non-physical variables. Because shade presence is a key factor influencing T_{mrt} and T_g, combining one of these approaches with ours would almost certainly reduce our model error.

Conclusions

Our findings reinforce the six-directional method as a critical tool for collecting reliable T_{mrt} data while demonstrating how the globe thermometer approach can be adapted for urban ecology research. Fine-scale T_{mrt} data provide scientists across disciplines with information about the influence of urban landscape heterogeneity on biophysical outcomes—critical information for cities striving to adapt to challenges such as densification and extreme weather conditions.

Although precise T_{mrt} measurements currently require specialized six-directional platforms like MaRTa, the accuracy of the globe thermometer method can be improved. Modifying the ISO Standard 7726 equation and averaging clusters of measurement points within a study area can reduce error. Further data are needed to evaluate the efficacy of alternative equations under diverse environmental conditions. However, our results suggest a simple improvement may lie in incorporating a low, constant V_a into the conventional equation, as proposed by Lin and Matzarakis (2011).

Researchers with access to six-directional T_{mrt} platforms should consider leveraging machine learning to develop localized adaptations of the globe thermometer method. However, ongoing validation against six-directional measurements would be essential, and models could not be applied beyond the range of conditions in which they were trained.

A third promising technique has recently emerged that utilizes small cylinders to estimate T_{mrt}. Developed by Rykaczewski et al. (2024), this cost-effective approach builds on the three-globe radiometer-anemometer concept developed by Nakayoshi et al. (2014), measuring shortwave and longwave radiation fluxes and the convection coefficient to estimate T_{mrt}. While this technique shows promise as an affordable and accurate alternative to both six-directional and globe thermometer

methods, it is still undergoing refinement (Rykaczewski et al. 2025) and was not suitable for application in our study.

As cities grow more complex and climate pressures intensify, the demand has never been greater for scientists to produce results that are both adaptable and reliable at hyper-local scales. Our work bridges methodological rigor with pragmatic field applications, carving a path forward for urban ecologists seeking to capture spatial and thermal complexity of urban landscapes. Rather than choosing between accuracy and efficiency, we argue for an integrative approach—where methodological innovation, like machine learning and spatial aggregation, enhances the accessibility and ecological relevance of T_{mrt} measurement. By refining the methods we use to sense and study the urban landscape, we sharpen our ability to ask and answer more meaningful questions about how humans and other organisms experience cities.

Declarations

Declaration of Funding

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Data Availability

Data will be made available upon request.

Competing Interests

The authors declare no competing interests.

Author Contributions

Conceptualization: Aubrey Benson, Melissa McHale. Methodology: Aubrey Benson, John Frank, Melissa McHale. Data collection: Aubrey Benson, John Frank, George Valentine. Analysis: Aubrey Benson, John Frank, George Valentine, Melissa McHale. Writing-original draft: Aubrey Benson. Writing-review and editing: Ben Crawford, John Frank, Ariane Middel, George Valentine, Travis Warziniack, Melissa McHale.

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Figures

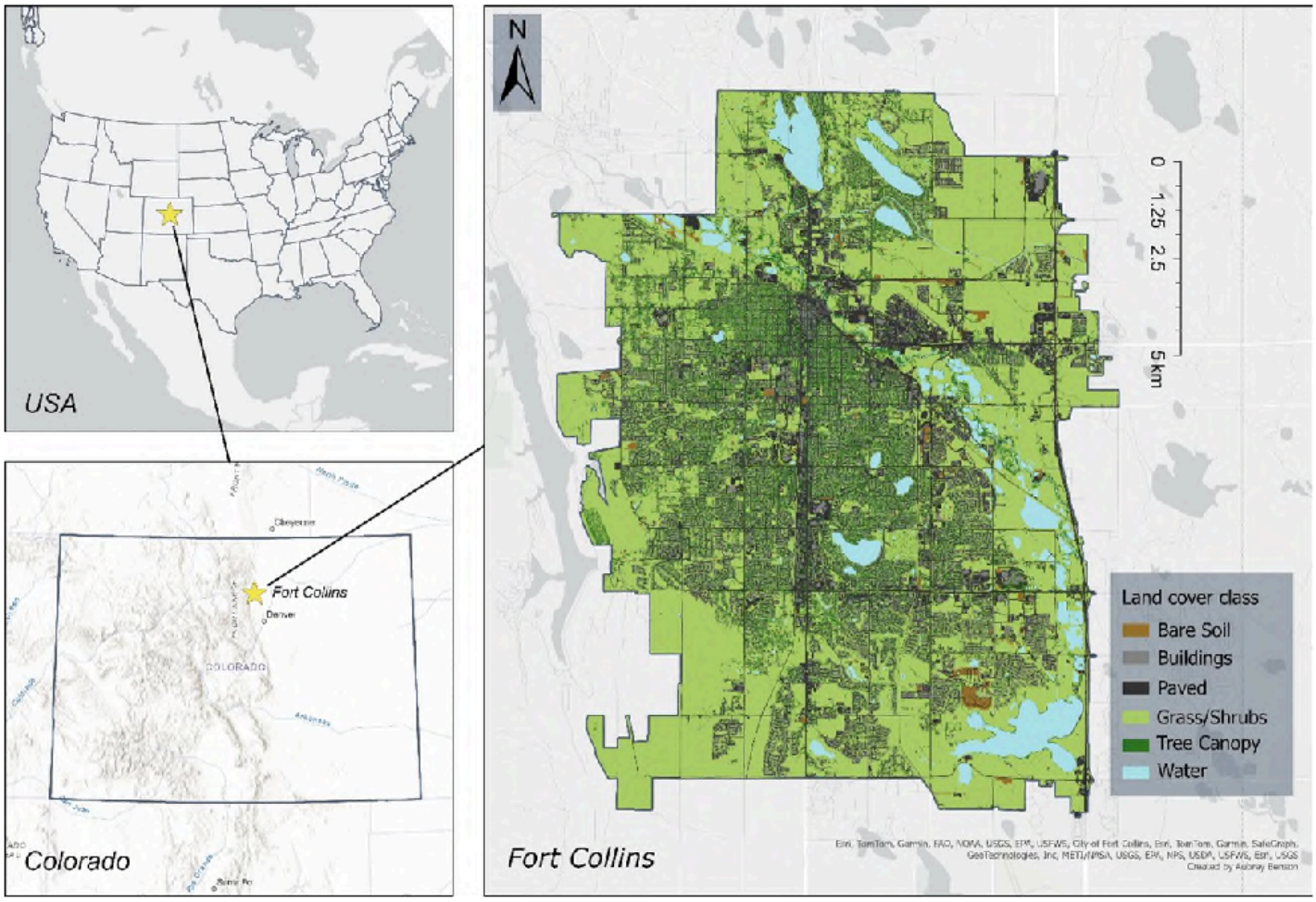


Figure 1

Fort Collins is a city in northern Colorado, USA, at the base of the Rocky Mountains.

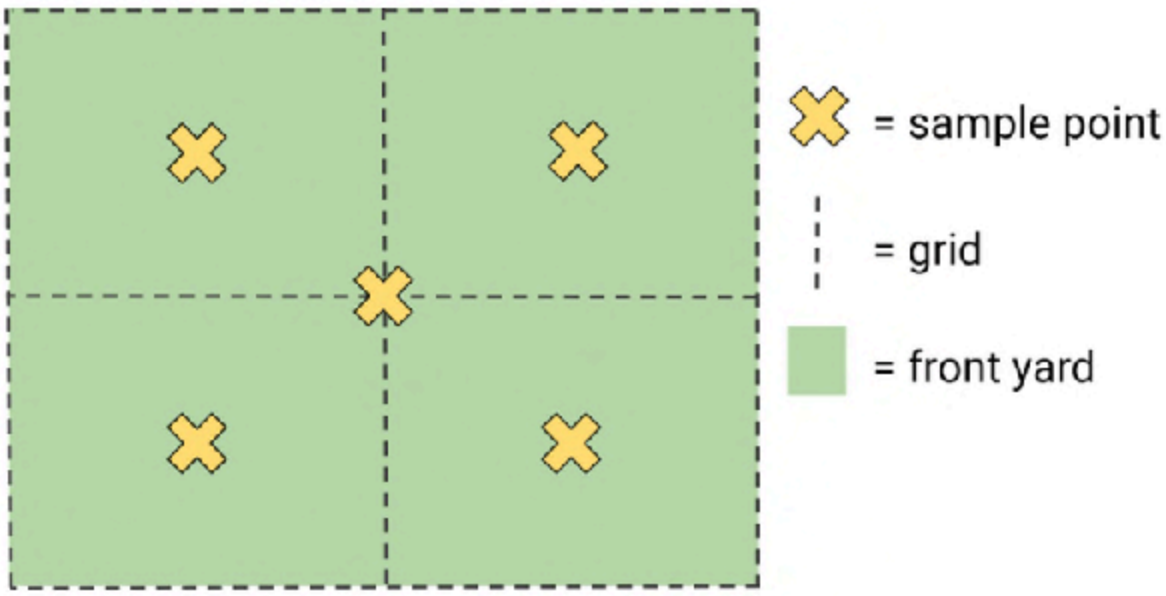


Figure 2

Systematic sampling design for the front yards studied. A two-by-two grid was first established, followed by placing five sample points. For the second year of data collection, in 2024, only four points were measured, omitting the central point in each yard.



Figure 3

The MaRTa machine (right) and Kestrel (left) were positioned opposite each other over each sample point (central yellow/orange “x”) and recorded data for one minute.



Figure 4

The MaRTa machine is equipped with shortwave and longwave radiation sensors in six directions and sensors for air temperature, surface temperature, relative humidity, and wind velocity.

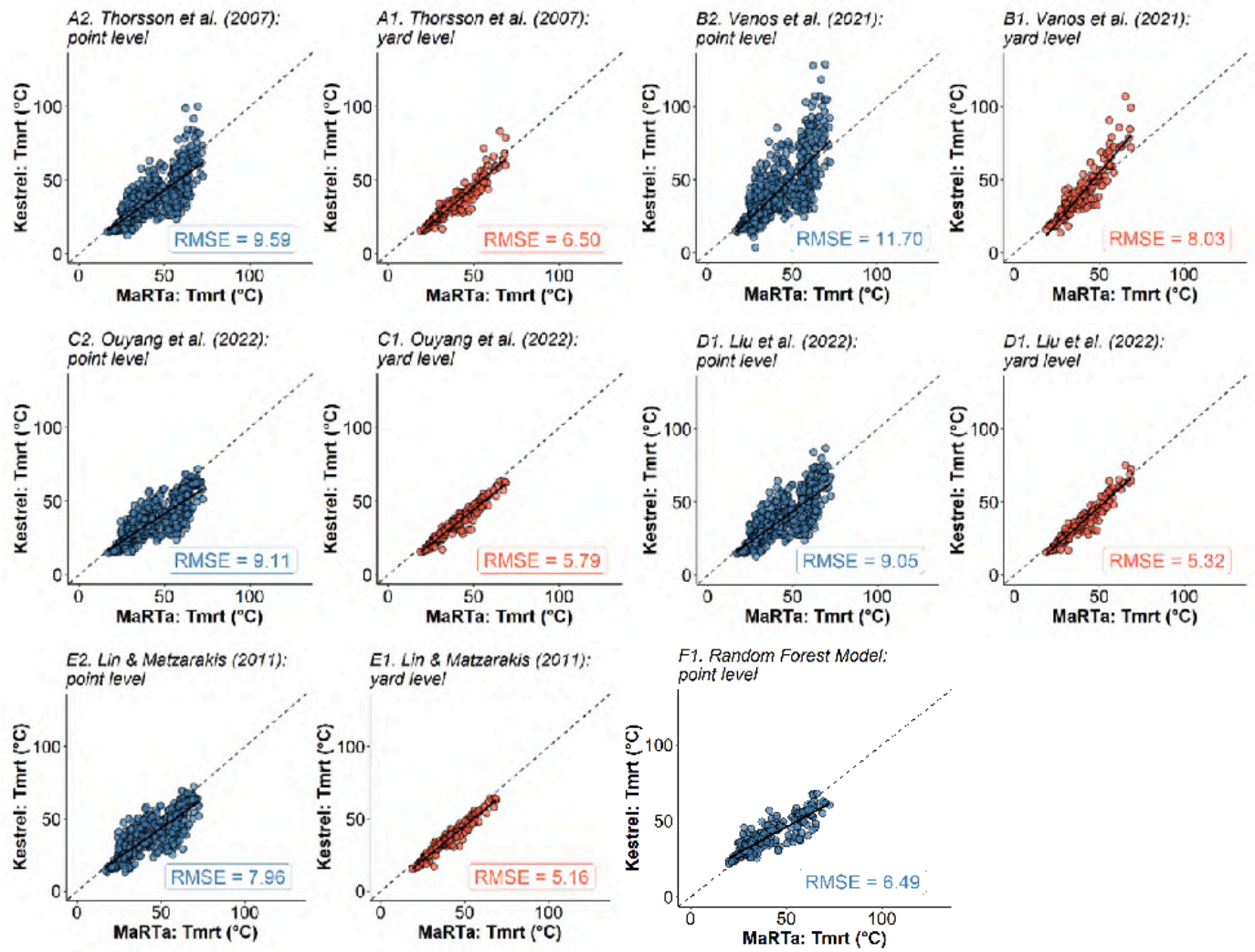


Figure 5

Comparison of Kestrel (globe thermometer) methods for calculating mean radiant temperature (Tmrt), compared to MaRTa (six-directional). Blue panels depict all points from all yards included in the study, while orange panels depict points aggregated at the front yard level. Solid black lines denote linear lines of best fit. Black dashed lines denote a theoretical slope of 1.

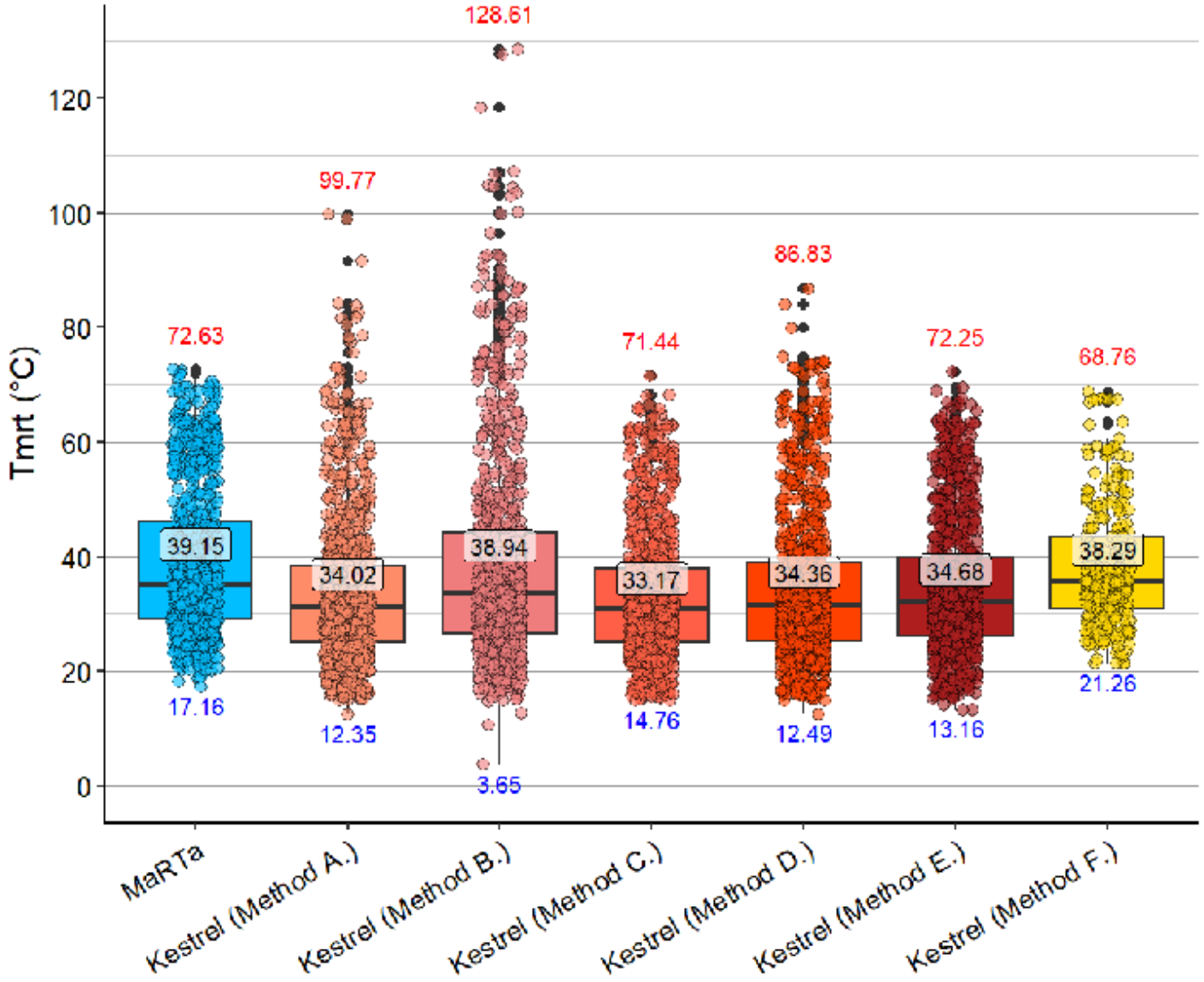


Figure 6

Comparison of Tmrt methods, including MaRTa (6-directional), (A.) conventional method, (B.) Vanos et al. (2021), (C.) Ouyang et al. (2022), (D.) Liu et al. (2022), (E.) Lin and Matzarakis (2011), and (F.) a random forest model. Red labels indicate maximum values for each dataset; blue labels indicate minimum values; black labels represent means.

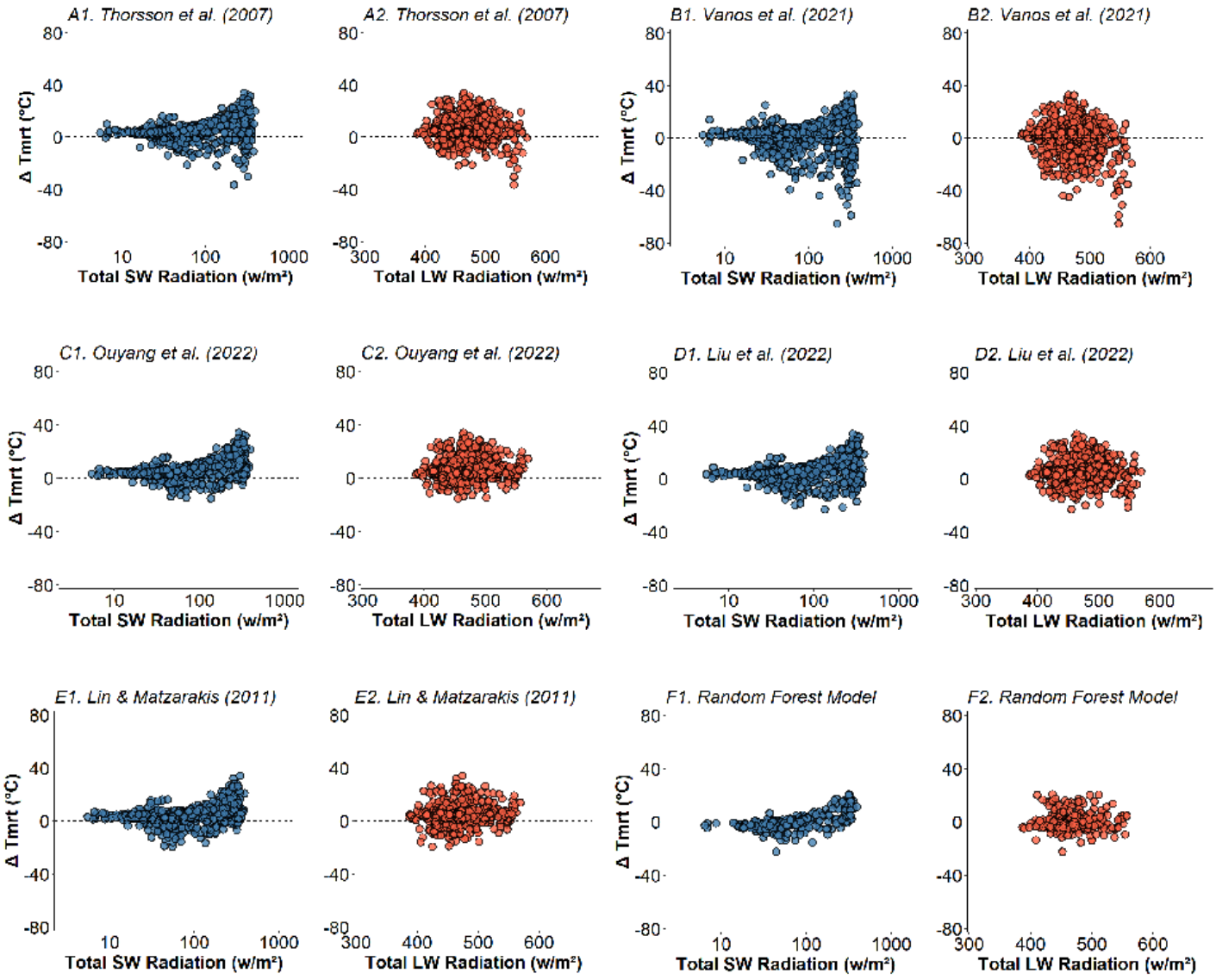


Figure 7

Relationships between radiation and the difference between Kestrel and MaRTa mean radiant temperature (ΔTmrt =MaRTa Tmrt-Kestrel Tmrt) values for multiple Kestrel T_mrt methods. Blue panels depict total incoming shortwave (SW) radiation, while orange panels depict total incoming longwave (LW) radiation. Radiation is weighted according to the six-directional method T_mrt equation (0.06 weighting for upward and downward radiation; 0.22 for lateral radiation). Note that shortwave panels use a semi-log scale.